

Design and Implementation of Electronic Circuitary for the Tilttable Microscope

Sachin kumar (23110286), Prof. Chandan Kumar Mishra

ABSTRACT - Due to the increasing application of colloids in various fields like medicine, material science, environmental science, etc, it is important to understand the behavior and characteristics of colloidal particles. This project aims to develop a tilting mechanism for a microscope to enable detailed observation of such behavior. The objective was to achieve precise angular positioning of the microscope using a sensor-based control system and to implement the design on a PCB. A DC motor controlled by an ESP32 microcontroller and an L293D driver is used to tilt the microscope. The BNO055 IMU sensor provides real-time pitch angle feedback. PWM signals are adjusted dynamically to control motor speed and direction based on feedback. The results shows smooth and controlled tilting, achieving a precision of less than 1° , supported by the effective braking mechanism as well. However, the targeted precision of less than 0.1° and PCB implementation could not be fully achieved due to time and design limitations. Nonetheless, the current prototype lays a strong foundation for future improvements in accuracy and integration.

INTRODUCTION

Typical microscopes operate in such a way that the microscope's objective and imaging plane are orthogonal to each other. While such a design has been useful in many cases, it precludes investigation into phenomena such as sedimentation. Confocal microscopy, in principle, can allow for studying phenomena like sedimentation, but its image acquisition rate is a major limitation. To overcome these challenges, one needs to design a microscope that can tilt, keeping the relative orientation of the imaging plane to the objective. The current prototype of a microscope in the experimental complex fluids lab at IITGN has been designed and implemented for the same. It can rotate to the target angle with a high precision of 0.1° .

OBJECTIVE

The objective of this project is to develop a motorized tilting system that allows the microscope to be tilted to a desired angle with resolution of less than 0.1° . The current mechanism is working fine but there is a room for improvements in it. We need a better rotating mechanism which can work effectively for longer time period. After this, it could get implemented on a PCB board.

PROBLEM STATEMENT

For better working and high efficiency, we need four basic operations to be performed:

1. A mechanism to rotate the microscope smoothly even at very small angular velocity of less than one rpm.
2. Feedback to ensure that the microscope has been rotated to the target angle correctly.
3. A safety measure to ensure that it does not rotate beyond the safe range.
4. A mechanism to stabilize the position of the microscope to prevent any sort of vibrations or fluctuations that might affect the readings.

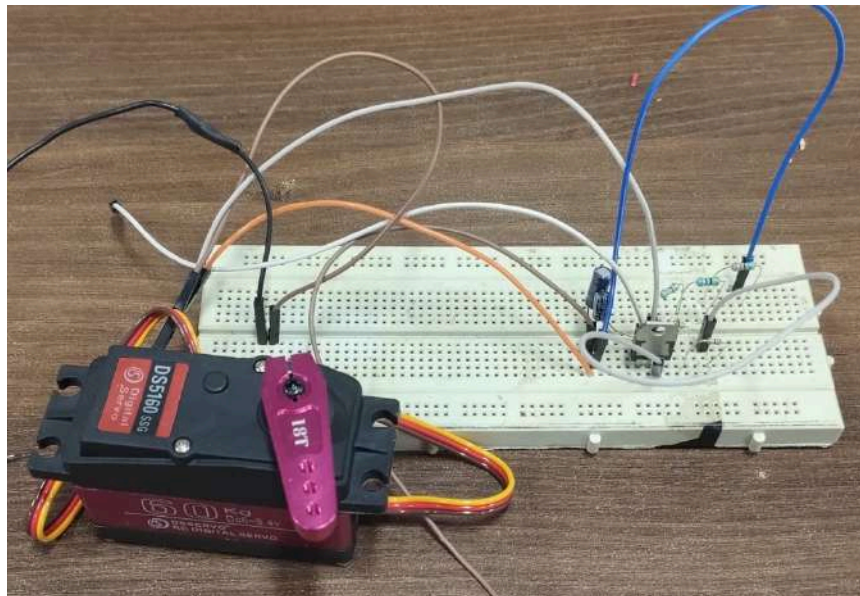
CURRENT MECHANISM

1. **Rotation:** For rotating the microscope, a DC (direct current) motor rated 12V, has been installed. Since its angular velocity is around 2500 rpm, a gear box of gear ratio 1000:1 has been fixed in series with the motor shaft to reduce the angular velocity of microscope below 1 rpm.
2. **Feedback:** For the feedback, a smartphone (with BOSCH motion sensors and Digital Motion Processor (DMP) unit) has been fixed on the top of the microscope which can directly give the real time angle rotated by the motor with a precision of 0.1° . An app has been developed to send this data to the CPU.
3. **Safety:** For the safety measures, a mercury switch and a magnetic sensor has been used. When the microscope rotates beyond 90° , the mercury inside the switch moves and completes the circuit of the safety signal which ensures that no further movement of the microscope takes place after this.
4. **Stabilization:** Currently a mechanical brake is being used for the stabilization. A compressor is used which compresses the air and creates a pressure difference. This pressure difference applies force on the valves which eventually results in pulling the strings of the brakes.

NEW MECHANISM

1. **Feedback:** For the feedback, we are using a BOSCH orientation sensor BNO055 which has inbuilt sensor fusion of sensors like gyroscope, magnetometer and accelerometer which gives the real time angle rotated by the microscope with a precision of 0.01° .

2. **Safety:** I have used the digital switch for this. I have modified the code of the ESP32 microcontroller and set the limit of angle from -80° to $+80^{\circ}$. The motor automatically stops if it crosses this limit.
3. **Braking mechanism :** We are required to reduce any sort of fluctuations or vibrations that can affect our readings. So, we need to stabilize our microscope at a fixed position. For this, we are applying an external braking mechanism to our setup that can pull a string and apply brakes when required.



Components :

1. Servo motor with a torque rating of 60 kgcm, which can pull the string easily.
2. ESP32 microcontroller to control the motor.
3. Since the voltage rating of the motor is 6 to 8.4 V, and we have a power supply of 12 V, we need a voltage regulator LM3171, some resistors, and capacitors to reduce the voltage to the desired range. In my case, the output voltage is 8.1 V.

Implementation :

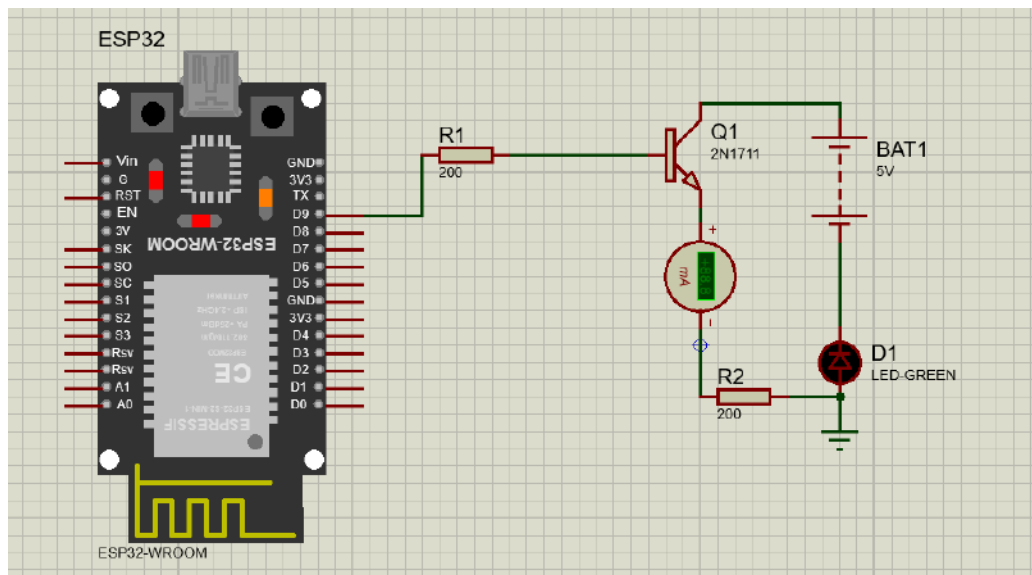
The string is connected to the servo horn. Whenever the input 'brake' is pressed, the horn rotates 90° in a clockwise direction. The length of the horn and its distance from the brake are set in such a way that when the motor rotates 90° , it pulls the string 4 cm in the forward direction. This results in applying the brakes smoothly. Again, when the input 'release' is entered, the motor rotates in a counterclockwise direction by 90° , and the brake returns to its actual position.

4. METHODOLOGY (ROTATION MECHANISM)

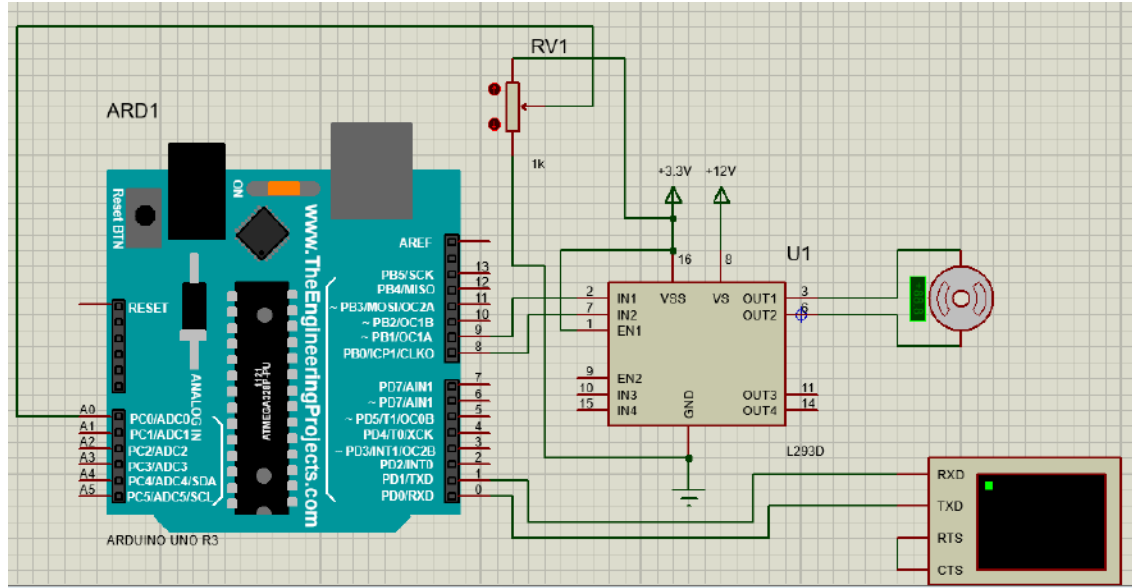
To rotate the microscope, we have decided to use the BLDC (brushless direct current) motor because it can run at very small angular velocity with good control, providing smooth rotation. It can also produce more torque. On the other hand, stepper motors move in steps so they can create small vibrations. If we try to reduce the step size to make the movement smoother, its load-bearing capability will decrease. A DC (direct current) motor again has less control at a smaller angular velocity, especially when the motor is under load. The contact between carbon brushes and the commutator causes mechanical friction, which limits its efficiency and lifespan.

To achieve these operations, rather than directly working upon hardware, firstly I decided to start with doing simulations on Matlab. I did some basic simulations, like blinking of an LED (light-emitting diode) and rotation of DC motors on Matlab. However, these simulations required the use of hardware. If hardware is required for conducting simulations, then the purpose of performing simulations is not fulfilled. So, to achieve this, I started to simulate on a more advanced simulating platform called Proteus. I simulated several operations on it.

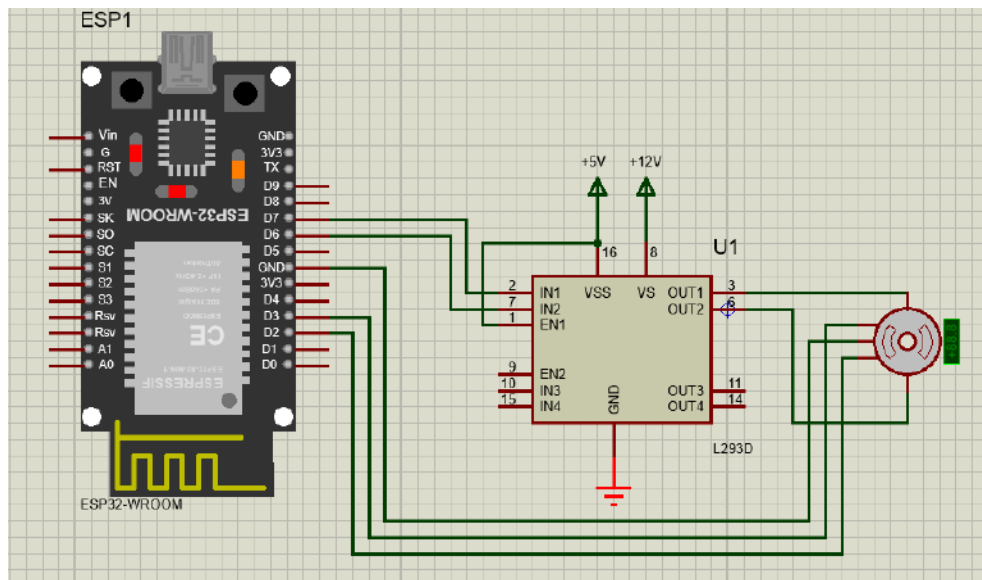
1. Firstly, I started with the basics and performed the same operation, blinking of an LED.



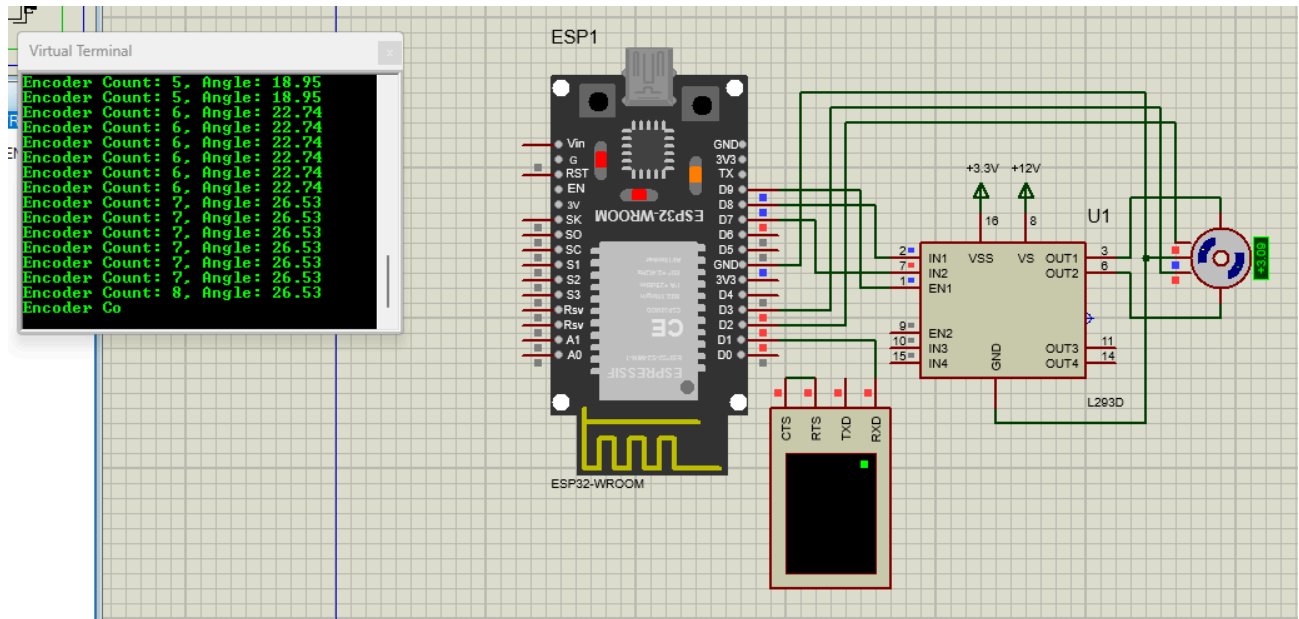
2. The Proteus does not support BLDC motors in their libraries. So I simulated a simple DC motor. I varied its angular velocity and also controlled its direction of motion.



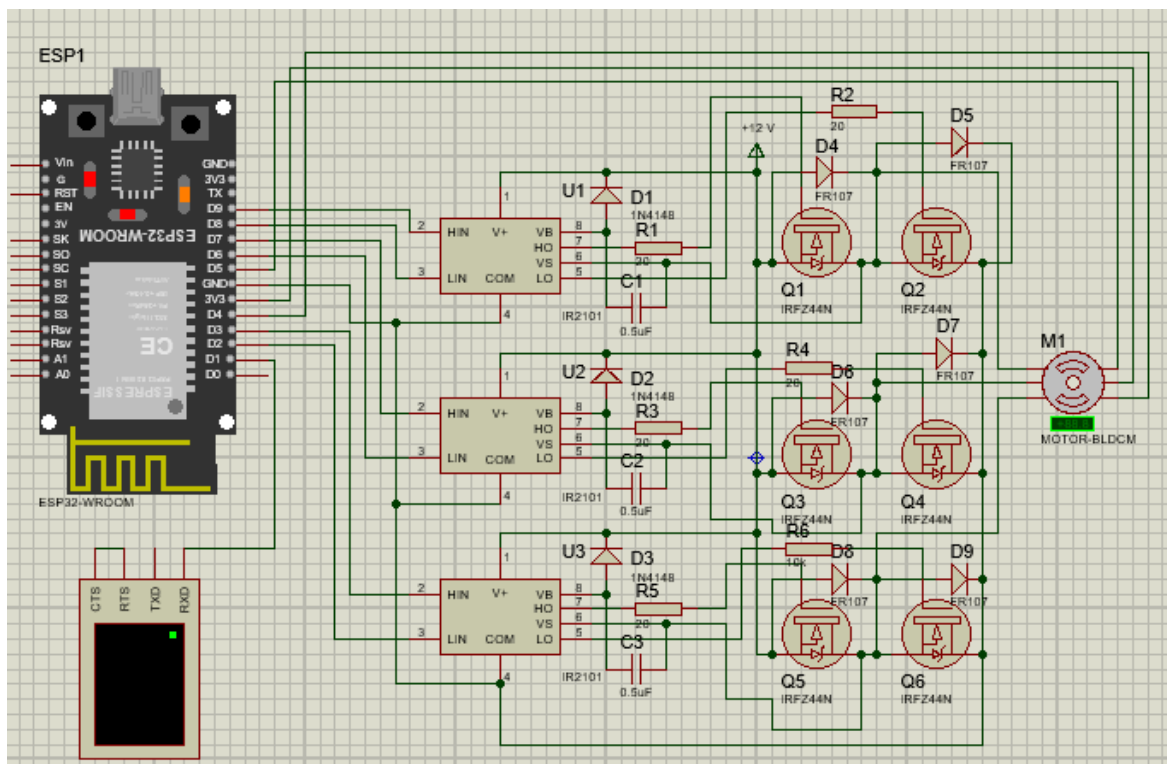
- Now, to measure the angle of rotation, I used an encoder DC motor. It detects the angle of rotation by generating the pulses whenever the shaft of the motor crosses a magnetic pole. Its simulation was also successful. But the step size of angle measurement in the encoder is around 3.79 degrees, which results in a very low resolution. We cannot get the desired angle correctly, which will result in false readings.



- After this, to apply the brakes, I compared the current angle rotated by the motor with the desired angle and started decelerating the motor's angular velocity 15° prior to the desired angle so that it can stop exactly at the desired angle smoothly.



- I tried to do all these operations on the BLDC motor as well. Since the BLDC motor library is not supported by Proteus, I designed the circuitry for this using three H-bridges for the three phases of the BLDC motor. But the position tracking in the BLDC motor is done through Hall sensors. Since the BLDC motor has 6 magnetic poles, It could measure the position only 6 times in the entire rotation. It was equivalent to a resolution of 60 degrees. This was even worse.



After doing all the simulations, now my objective was to test it physically. I started working on the BLDC motor.

Components :

1. BLDC motor to rotate the setup.
2. Gearbox to reduce the angular velocity of the motor.
3. BNO055 orientation sensor to measure the real-time position of the motor.
4. ESC (Electronic Speed Controller) to control the BLDC motor.
5. Relay modules to reverse the direction of the motor.
6. ESP32 microcontroller to control the PWM (pulse width modulation) signals.
7. 12 V power supply for motor and 5 V power supply for microcontroller.

It was rotating smoothly. Its direction and angular velocity could be controlled. I tested it by rotating the angle sensor manually to the target angle, and it was actually stopping at the target angle. It was working in both directions with ease, but it had some limitations as well.

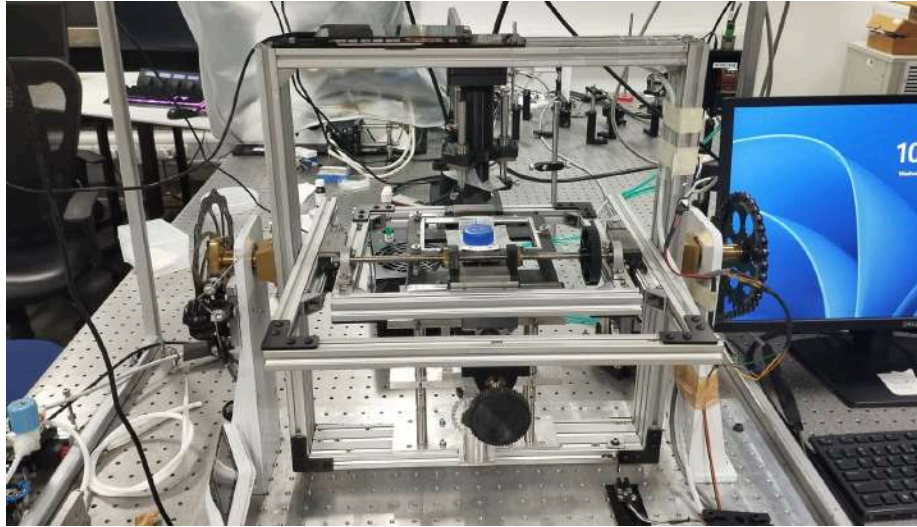
1. Generally the BLDC motors are manufactured for high angular velocity operations. Its angular velocity range was from 4000 to 10000 rpm. It could not be controlled below 4000 rpm. It means that its angular velocity was decreasing from 4000 rpm to directly 0 rpm. Even after using the 1000:1 gearbox, its minimum angular velocity would be 4 rpm. Its angular velocity would decrease from 4 to directly 0 rpm, which could prevent us from getting the smooth rotation while stopping.
2. To further reduce the angular velocity of the motor, we could not apply two gearboxes in series with the motor. As the gear ratio increases, the current rating also increases. A high current rating makes the motor more prone to damages. So, it was not a good solution.

We have searched on various websites and contacted various manufacturers nearby. But we could not find any specific motor with all the characteristics that we needed. If any motor had good control at low rpm, then it did not have higher torque capability. If any motor had better torque capability, then its size was large and power ratings were high.

So, to achieve our objective, we went back to the stepper motor. Stepper motor has better position control, and its step size can also be decreased for smooth rotation. But it also had some limitations.

1. Stepper motors move in steps so they can create small vibrations.
2. At small angular velocity and large torque, it starts missing its steps.
3. If we try to reduce its step size to make the movement smoother, its load bearing capability will decrease.
4. It might result in jerks, overheating, less accuracy and shorter lifespan.

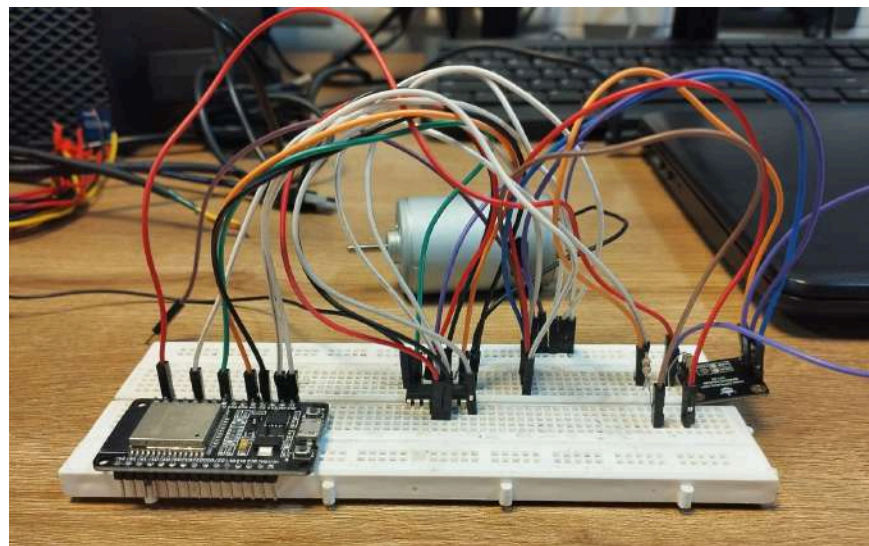
SYSTEM DESIGN AND IMPLEMENTATION



This is the physical setup of the microscope, which is rotated using the DC motor. Though the DC motor has some limitations, we can control it at a much smaller speed than the BLDC motor. We can control its PWM and apply a gearbox to reduce its speed to the desirable range.

Components :

- A 12 V DC motor has been used in this design to tilt the microscope.
- A gearbox of gear ratio 1000:1 is used to reduce the angular velocity of the motor.
- Angle sensor BNO055 determines the real-time position of the motor.
- L293D motor driver to control the angular velocity and direction of the motor.
- ESP32 microcontroller to perform all the operations.
- 12 V power supply for motor and 5 V power supply for microcontroller.



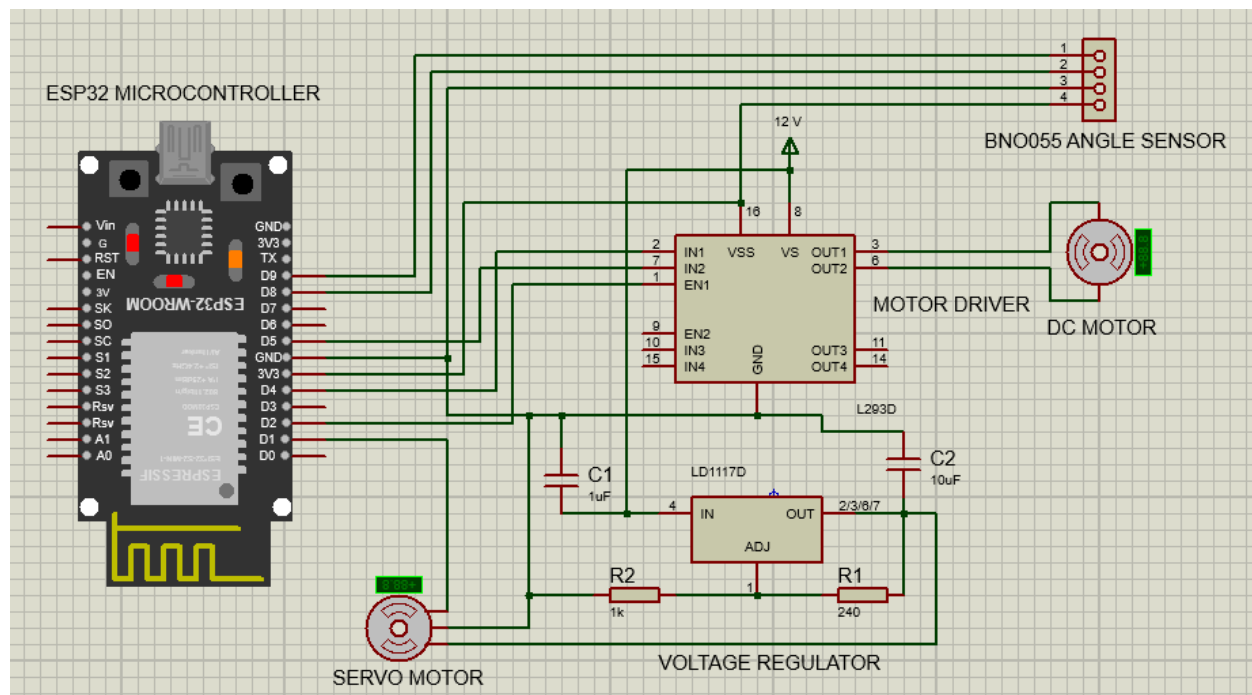
This whole setup shown in the picture is fixed upon the microscope. Whenever the motor rotates, the microscope along with the whole setup rotates. Since the orientation of the BNO055 changes, the sensors inside it (gyroscope, accelerometer, and magnetometer) sense this change with the resolution of 0.1° . It gives the data of the real-time position of the motor to the microcontroller.

When I give the target angle as an input to the ESP32, it compares it with the readings of BNO055 (real time angular position of the motor). If the target angle is more than the actual angle of the motor then the motor rotates in clockwise direction. If the target angle is less than the actual angle then the motor driver, which consists of H-bridge, changes the direction of the motor and the motor rotates in counterclockwise direction.

The angular velocity of the motor is controlled by the PWM signals. Since it consists of 8 bits, its value ranges from 0 to 255. 0 determines 0% duty cycle, and 255 determines 100% duty cycle. At PWM value 0, the motor's angular velocity is 0, and at 255, it is around 4000 rpm. The values of angular velocity depend upon the frequency of the PWM as well. In my case, it is 1000 Hz.

But it does not mean that the motor starts rotating slowly at PWM value 1. Due to the very small duty cycle, the average voltage experienced by the motor is too low at small values of PWM. It starts rotating at PWM value 90, which is equivalent to around 200 rpm. We cannot control the motor below this rpm. It has the same problem as in a BLDC motor, but in this case the range of uncontrolled angular velocity is very low.

This is the complete circuit diagram of the design:



To start and stop the motor smoothly, I have fixed three different values of PWM to control the angular velocity of the motor. The start PWM, maximum PWM, and minimum PWM. The start PWM is greater than the minimum PWM because it takes a large initial PWM to break the inertia and rotate the motor. But during stopping, due to inertia, it remains rotating even at smaller PWM values.

It accelerates smoothly from the start PWM to maximum PWM for the first 1/3 of the distance. For the remaining 2/3 of the distance, it decelerates slowly from the maximum PWM to the minimum PWM; i.e., if the target angle is 50° and the current position is 20° , then it accelerates from 20° to 30° and then decelerates from 30° to 50° . Since the angular velocity of the motor ranges from 200 to 2200 rpm, the gearbox has been attached, which reduces its angular velocity to less than 1 rpm.

There is some limitation in the angle sensor BNO055 as well. Though it is rated to calculate angles from -90° to $+90^\circ$, on using it physically, I found that it can give angles from -87° to $+87^\circ$ only. Means if the target angle is 90° then it will never find that angle and will be rotating forever. To fix this issue, I have set its rotating range from -80° to $+80^\circ$. So, whenever the target angle is more than $+80^\circ$ or below -80° , then the new target angle becomes $+80^\circ$ or -80° itself. For further safety limits, the motor will automatically stop if it crosses $+82^\circ$ or -82° .

RESULTS

My design was working. I was able to rotate the microscope using the DC motor at a small angular velocity. I could control its position and stop it at the desired angle smoothly. But it was not perfect. Initially, it had an overshoot of 1.5° in the clockwise direction and 0.5° in the counterclockwise direction. I observed that the behavior of the motor was not the same in both directions. The overshoot was also varying for different target angles. So, I made different adjustments of the starting, maximum, and minimum PWMs for different target angles in both directions. And, actually, it worked. The overshoot now decreased to 0.8° in the clockwise direction and 0.3° in the counterclockwise direction.

CONCLUSION

The objective to rotate the microscope at an angle with a precision of less than 0.1° was not fulfilled. My design could achieve the resolution of 0.3° to 0.8° only. Therefore, I could not implement it on the PCB. The feedback mechanism is working fine by placing the angle sensor BNO055 on the top of the microscope, but it too has some calibration errors.

The safety measure had been successfully applied. The motor stops automatically if it crosses the safety range. The braking mechanism is also working as per the expectations. It can easily pull the brakes after reaching the target angle. But I could not get the chance to apply it on the physical setup.

FUTURE WORK

- The overshoot of the microscope can be reduced by two methods. First is by varying the PWM frequency and torque of the motor. Second is by making different adjustments of the starting, maximum, and minimum PWMs for different target angles. It might demand much more effort and patience to get the desired resolution and efficiency, which could not be achieved by me in the remaining one week.
- Once the overshoot is reduced, the BNO055 can be calibrated easily to the desired angle to reduce the calibration errors.
- The braking mechanism is working fine. It can be implemented in the setup. It might encounter some issues as well. But it can also be fixed by doing experiments.